# Homework 1

MATH 518 - Fall 2025

**Due on Friday, September 26, 11:59pm on Crowdmark**

Total of points = 75 points

Please submit each question separately on Crowdmark. Unless stated otherwise, $k$ is an algebraically closed field of characteristic 0.

1. (a) (5 points) Show that any prime ideal is radical.

   (b) (5 points) Let $I$ be an ideal in $k[x_1, \ldots, x_n]$. Show that $V(I) = V(\mathrm{rad}(I))$.

   (c) (5 points) Let $V$ be an algebraic set. Show that $I(V)$ is a radical ideal.

**Solution:**

(a) Let $I$ be a prime ideal. We know that $I \subset \mathrm{rad}(I)$ and want to show that the equality holds. Suppose we have an element $f \in \mathrm{rad}(I) \setminus I$. Then $f^k = f \cdot f^{k-1} \in I$ for some $k > 1$. Since $I$ is prime, we deduce that $f \in I$ and get a contradiction.

(b) Since $I \subset \mathrm{rad}(I)$, we have $V(\mathrm{rad}(I)) \subset V(I)$. On the other hand, if $p \in I$ then $f(p) = 0$ for all $f \in I$. Thus if $f \in \mathrm{rad}(I)$, we have $f^k(p) = (f(p))^k = 0$. We deduce that $p \in V(\mathrm{rad}(I))$.

(c) Suppose that $f \in \mathrm{rad}(I(V))$. Then $f^r \in I(V)$ for some $r \geq 0$. This means that for all $p \in V$, we have $f^r(p) = 0$. This implies that $f(p) = 0$ for all $p \in V$, and we deduce that $\mathrm{rad}(I(V)) \subset I(V)$.

2. (a) (5 points) Let $I, J$ be two coprime[1] ideals in a commutative ring $R$. Show that $IJ = I \cap J$.

   (b) (5 points) Write generators of the ideal $I(V) \subset k[x, y]$ where

$$V = V(xy^2, (x-1)(y+1)^2) \subset k^2.$$

[1] Two ideals are coprime if $I + J = R$.

**Solution:**

(a) We always have $IJ \subset I \cap J$. On the other hand, if $I + J = R$ then we can find $a \in I$ and $b \in J$ such that $a + b = 1$. Thus for any $x \in I \cap J$ we have $x = xa + xb \in IJ$.

(b) We have

$$\begin{aligned} V(xy^2, (x-1)(y+1)^2) &= V(xy, (x-1)(y+1)) \\ &= V(x, y+1) \cup V(x-1, y) \\ &= \{(0,-1)\} \cup \{(1,0)\}. \end{aligned}$$

The ideal $(x, y+1)$ and $(x-1, y)$ are coprime, so that $(x, y+1) \cap (x-1, y) = (x, y+1)(x-1, y) = (x(x-1), y(y+1), xy, (y+1)(y-1))$. We conclude by recalling that $I(V \cup W) = I(V) \cap I(W)$ for any two algebraic sets $V, W$, so that

$$\begin{aligned} I(V) &= (x, y+1) \cap (x-1, y) \\ &= (x, y+1)(x-1, y) \\ &= (x(x-1), y(y+1), xy, (y+1)(y-1)). \end{aligned}$$

3. (10 points) The Zariski closure of a set $W \subset k^n$ is the smallest Zariski closed subset $\overline{W} \subset k^n$ such that $W \subset \overline{W}$. Show that

$$\overline{W} = \bigcap_{\substack{W \subset Y \\ Y \text{ is closed}}} Y = V(I(W)).$$

**Solution:** Since $\bigcap_{\substack{W \subset Y \\ Y \text{ is closed}}} Y$ is closed and contains $W$, it is clear that

$$\overline{W} \subset \bigcap_{\substack{W \subset Y \\ Y \text{ is closed}}} Y. \tag{1}$$

On the other hand, since $V(I(W))$ is a closed subset containing $W$, it appears as one of the $Y$'s in the RHS. So we have

$$\bigcap_{\substack{W \subset Y \\ Y \text{ is closed}}} Y \subset V(I(W)). \tag{2}$$

Finally, since $W \subset \overline{W}$, we have $I(\overline{W}) \subset I(W)$ and thus $V(I(W)) \subset V(I(\overline{W})) = \overline{W}$.

4. For each of the following algebraic subsets, find the unique decomposition into irreducibles.
   (a) (5 points) $V(y - x^2) \subset k^2$.
   (b) (5 points) $V(x^2 + y^2 - 1, x^2 - z^2 - 1) \subset k^3$.
   (c) (5 points) $V(y^4 - x^2, y^4 - x^2y^2 + xy^2 - x^3) \subset k^2$.
   (d) (5 points) In each case, are the irreducible components different if we take $k = \mathbb{R}$ (not algebraically closed)?

**Solution:**

(a) (5 points) The ideal $(y - x^2)$ is prime, since

$$k[x, y]/(y - x^2) \simeq k[x]$$

is an integral domain. Thus $V(y - x^2)$ is irreducible.

(b) (5 points) The first polynomial implies the condition $x^2 - 1 = -y^2$, so that

$$\begin{aligned}
&V(x^2 + y^2 - 1, x^2 - z^2 - 1) \\
&= V(x^2 + y^2 - 1, y^2 + z^2) \\
&= V(x^2 + y^2 - 1, y + iz) \cup V(x^2 + y^2 - 1, y - iz).
\end{aligned}$$

(Here $i$ denotes an element in $k$ such that $i^2 = -1$). Note that $x^2 + y^2 - 1$ is irreducible in $k[x, y]$, so that $(x^2 + y^2 - 1)$ is a prime ideal (in a UFD all irreducibles are prime). Thus

$$k[x, y, z]/(x^2 + y^2 - 1, y \pm iz) \simeq k[x, y]/(x^2 + y^2 - 1)$$

is an integral domain, and we deduce that the two component are irreducible. They are also not contained in each other.

(c) (5 points) Note that $y^4 - x^2y^2 + xy^2 - x^3 = (y^2 + x)(y - x)(y + x)$ and $y^4 - x^2 = (y^2 - x)(y^2 + x)$. Thus we have

$$\begin{aligned}
V &= V(y^4 - x^2, y^4 - x^2y^2 + xy^2 - x^3) \\
&= V(y^2 + x) \cup V(y^2 - x, y - x) \cup V(y^2 - x, y + x).
\end{aligned}$$

We have

$$\begin{aligned}
V(y^2 - x, y - x) &= V(x, y) \cup V(x - 1, y - 1) = \{(0, 0)\} \cup \{(1, 1)\} \\
V(y^2 - x, y + x) &= V(x, y) \cup V(x - 1, y + 1) = \{(0, 0)\} \cup \{(1, -1)\},
\end{aligned}$$

and so

$$V = V(y^2 + x) \cup \{(0, 0)\} \cup \{(1, 1)\} \cup \{(1, -1)\}.$$

All these terms are irreducible, but $\{(0, 0)\}$ is contained in $V(y^2 + x)$. Thus the unique decomposition is

$$V = V(y^2 + x) \cup \{(1, 1)\} \cup \{(1, -1)\}.$$

(d) In a) or c) the components are the same if $k = \mathbb{R}$. But in b), we ahve

$$V(x^2 + y^2 - 1, y^2 + z^2) = V(x^2 - 1, y, z) = \{(1, 0, 0)\} \cup \{(-1, 0, 0)\}.$$

5. (10 points) Let $V = \{(t, t^2, t^3) \in k^3 \mid t \in k\}$. Find generators of the ideal $I(V)$ and deduce that $V$ is an irreducible algebraic set in $k^3$.

**Solution:** Consider the ideal

$$I = (y - x^2, z - x^3).$$

It is clear that $V \subset V(I) = V(y - x^2) \cap V(z - x^3)$. Conversely, any point in $V(I) = V(y - x^2) \cap V(z - x^3)$ must be of the form $(x, x^2, x^3)$. Thus $V = V(I)$ is an algebraic set. The ideal $I$ is prime, since

$$k[x, y, z]/(y - x^2, z - x^3) \simeq k[x, y]/(y - x^2) \simeq k[x]$$

is an integral domain, so $V$ is an irreducible algebraic set. Moreover, $I$ is radical, and by the Nullstellensatz we find that

$$I(V) = I = (y - x^2, z - x^3). \tag{3}$$

6. Prove or provide a counterexample to the following statements.
   (a) (5 points) The intersection of two irreducibles is always irreducible.
   (b) (5 points) The Zariski topology on $k^{n+m}$ agrees with the product topology[2] on $k^n \times k^m$.

**Solution:**

(a) False. The set $V(x^2 + y^2 - 1)$ and $V(y)$ are both irreducible in $k^2$, but $V(x^2 + y^2 - 1) \cap V(y) = V(x^2 + y^2 - 1, y) = V(x^2 - 1) = \{-1, 1\}$ is reducible.

[2]The product topology on the product $X_1 \times X_2$ of two topological spaces is the topology with basis the sets of the form $U_1 \times U_2$, with $U_i$ open in $X_i$. In particular, it makes the projections $p_i \colon X_1 \times X_2 \to X_i$ continuous.

(b) False. Consider the Zariski closed set $V(xy-1)$ in $k^2$, which is the hyperbola $\{(x, x^{-1}) \in k^2 \mid x \neq 0\} \subset k^2$. If it was closed in the product topology, then image $\pi_1(V(xy-1)))$ should be closed (since $\pi_1$ is continous). But the image

$$\pi_1(V(xy-1))) = k \setminus \{0\} \tag{4}$$

is open in $k$.